CEE6501 — Lecture 9.2

3D Frame Analysis

Learning Objectives

By the end of this lecture, you will be able to:

Agenda

Part 1 —

Part 1 — DSM Element Roadmap

> *From trusses to 2D beams to 2D frames to 3D frames*

The 3D Frame Element (Where We Are Headed)

- A general 3D frame member has **6 DOFs per node** (12 per element)
  - translations: u_x, u_y, u_z
  - rotations: $\theta_x, \theta_y, \theta_z$
- This single element can represent **axial**, **torsion**, and **bending** behavior

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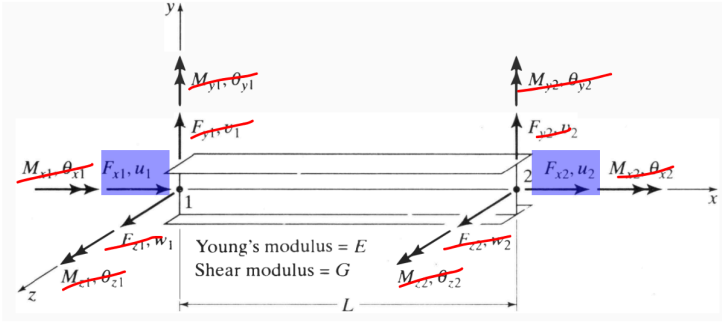
Key Idea: Superposition

You can think of the 3D frame element as a **superposition** of four behaviors:

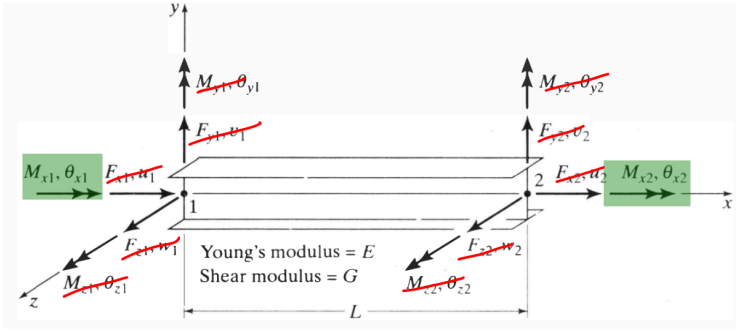
1. **Axial** deformation (truss-like)
2. **Torsion** about the member axis
3. **Bending** about one principal axis (Z-axis)
4. **Bending** about the other principal axis (Y-axis)

The Four Behaviors

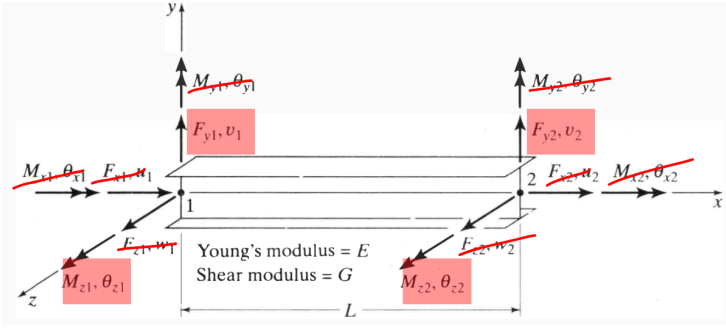
(1) Axial



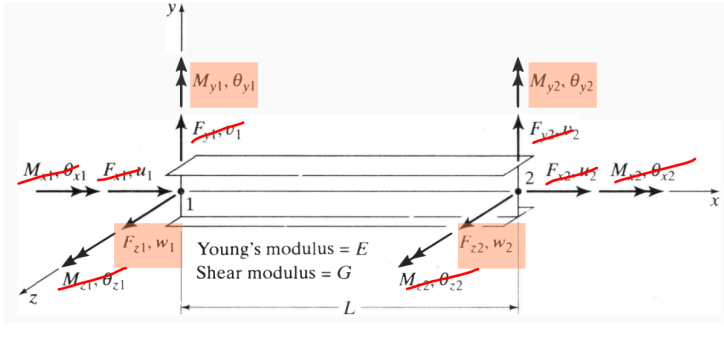
(2) Torsion



(3) Bending about Z-axis



(4) Bending about Y-axis



What We Derived Last Lecture

- The **2D beam element** governing planar bending behavior (cases (3) and (4))
- Once derived for one principal axis, the formulation extends directly to the other (identical mathematics, different axis)

What We Will Derive Today

- The DSM formulation for the **3D frame element**

Wrap-Up

In this lecture, we:

- Introduced the 2D frame analytical model and DOF conventions
- Reviewed local frame stiffness relations and the 2D frame stiffness matrix
- Worked through a complete example, including load resolution and fixed-end forces

Next lecture:

- Transforming local frame stiffness to global
- DSM for frame structures